HALF TWO · THE HARDWARE OF AI

The chips that make AI possible

Why energy — not intelligence — is the hardest problem in AI hardware.

SHASHANK BANSAL · ISN LAB · UC SAN DIEGO

UC San Diego
JACOBS SCHOOL OF ENGINEERING
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Everything you just heard is one operation, repeated

01 Embeddings, attention, multimodal fusion — all of it is numbers in, numbers out.

02 Learning those weights comes down to matrix multiplication, over and over.

03 Multiply-and-add, done a trillion times a second — that has to run on something physical.

So — where does all that math actually happen?

Time to open up the machine.

SECTION 01

From scrolling to switches

What's actually happening inside the chip in your pocket.

AUDIENCE POLL

A 10-second TikTok is made of how many 1s and 0s?

A

A few thousand

B

A few million

C

A few billion

B — tens of millions of bits

~30,000,000 bits

Every one of those bits is a 1 or a 0 — and something physical has to flip each one. That something is a transistor.

A transistor is just a switch

STEP 1 / 3

0 · OFF — NO CURRENT
1 · ON — CURRENT FLOWS

A tiny electrical gate with two states. Flip it, and the light turns on — that is the entire alphabet of computing.

A transistor is just a switch

STEP 1 / 3

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A tiny electrical gate with two states. Flip it, and the light turns on — that is the entire alphabet of computing.

Switches combine into logic

STEP 2 / 3

TWO SWITCHES IN SERIES · BULB LIGHTS ONLY WHEN BOTH ARE CLOSED · AN “AND” GATE

Wire two switches together and they start making decisions: the light only turns on when both agree. Logic, built from switches.

Switches combine into logic

STEP 2 / 3

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Logic builds arithmetic

STEP 3 / 3



Stack enough simple switches and you get the machine that runs ChatGPT.

Logic builds arithmetic

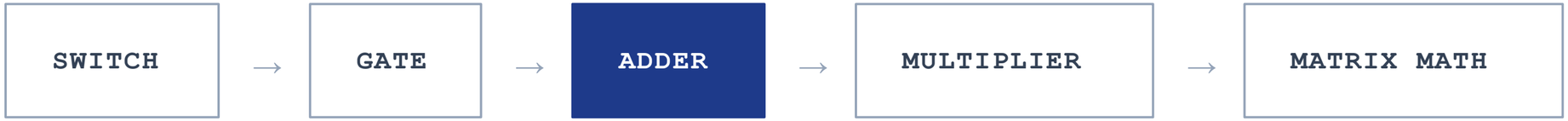
STEP 3 / 3



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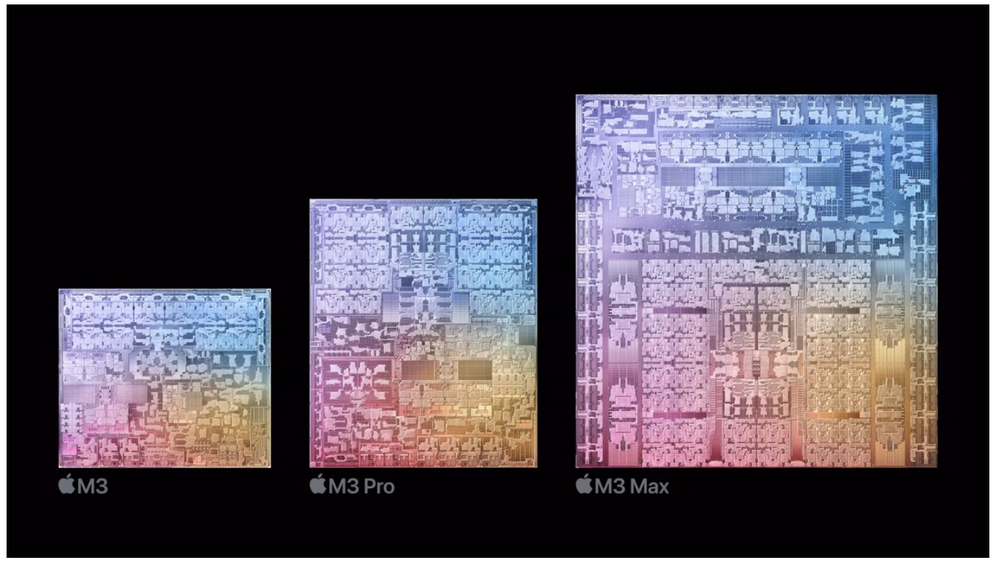
Logic builds arithmetic

STEP 3 / 3



Stack enough simple switches and you get the machine that runs ChatGPT.

Nineteen billion switches, on a chip the size of a fingernail



*https://appleinsider.com/articles/23/11/10/what-apples-three-gpu-enhancements-in-a17-pro-and-m3-actually-do

92,000,000,000

TRANSISTORS IN ONE M3 MAX CHIP

More than 12 switches for every human being on Earth — packed onto a sliver of silicon smaller than a fingernail.

SECTION 02

The energy wall

All this computing has a power bill — and it is getting steep.

Every word ChatGPT writes is billions of multiply-adds

Watch the running total climb, word by word.

The sky is blue today

0

RUNNING TOTAL · MULTIPLY-ADDS

This counts inference only — generating one word, not training the model. The ~2× multiplier is one multiply and one add per stored weight, per output word (Kaplan et al., "Scaling Laws for Neural Language Models," 2020).

Every word ChatGPT writes is billions of multiply-adds

Watch the running total climb, word by word.

+70B

The sky is blue today

70,000,000,000

RUNNING TOTAL · MULTIPLY-ADDS

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Every word ChatGPT writes is billions of multiply-adds

Watch the running total climb, word by word.

+70B +70B

The **sky** is blue today

140,000,000,000

RUNNING TOTAL · MULTIPLY-ADDS

This counts inference only — generating one word, not training the model. The ~2× multiplier is one multiply and one add per stored weight, per output word (Kaplan et al., "Scaling Laws for Neural Language Models," 2020).

Every word ChatGPT writes is billions of multiply-adds

Watch the running total climb, word by word.

+70B The +70B sky +70B is blue today

210,000,000,000

RUNNING TOTAL · MULTIPLY-ADDS

This counts inference only — generating one word, not training the model. The ~2× multiplier is one multiply and one add per stored weight, per output word (Kaplan et al., "Scaling Laws for Neural Language Models," 2020).

Every word ChatGPT writes is billions of multiply-adds

Watch the running total climb, word by word.

+70B The +70B sky +70B is +70B blue today

280,000,000,000

RUNNING TOTAL · MULTIPLY-ADDS

This counts inference only — generating one word, not training the model. The ~2× multiplier is one multiply and one add per stored weight, per output word (Kaplan et al., "Scaling Laws for Neural Language Models," 2020).

Every word ChatGPT writes is billions of multiply-adds

Watch the running total climb, word by word.

+70B	+70B	+70B	+70B	+70B
The	sky	is	blue	today

350,000,000,000

RUNNING TOTAL · MULTIPLY-ADDS

This counts inference only — generating one word, not training the model. The ~2× multiplier is one multiply and one add per stored weight, per output word (Kaplan et al., "Scaling Laws for Neural Language Models," 2020).

Memory and math live in different places

STEP 1 OF 3 · THE SETUP



The weights live in memory. The math happens somewhere else. This is how almost every computer is built.

Every multiply means a round trip

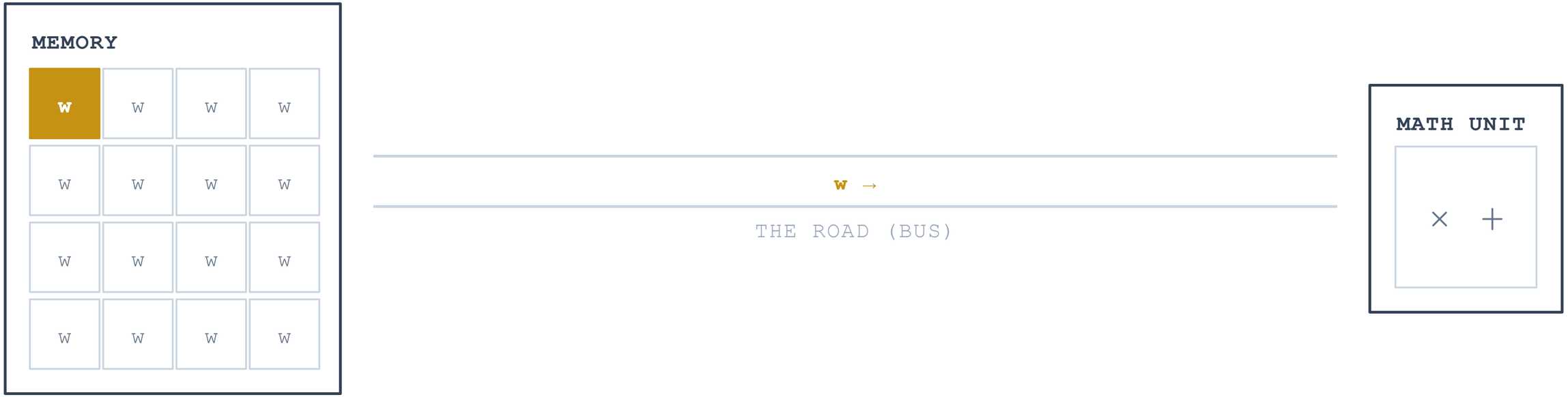
STEP 2 OF 3 · ONE TRIP



Pick one weight. To multiply it, it first has to leave memory.

Every multiply means a round trip

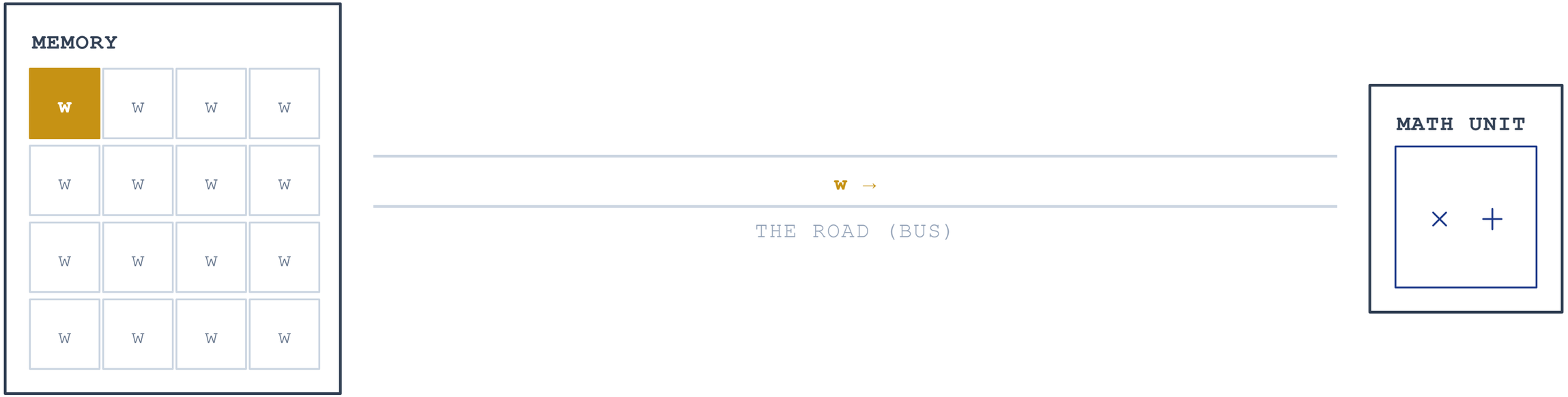
STEP 2 OF 3 · ONE TRIP



It travels across the chip to reach the math unit.

Every multiply means a round trip

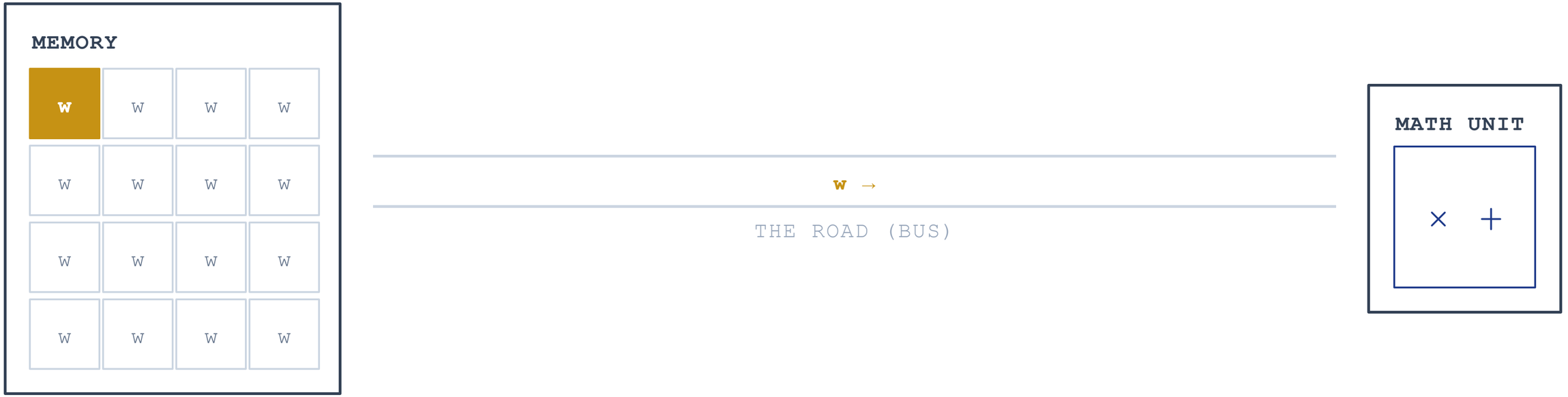
STEP 2 OF 3 · ONE TRIP



It arrives. Now — finally — the multiply happens.

Every multiply means a round trip

STEP 2 OF 3 · ONE TRIP



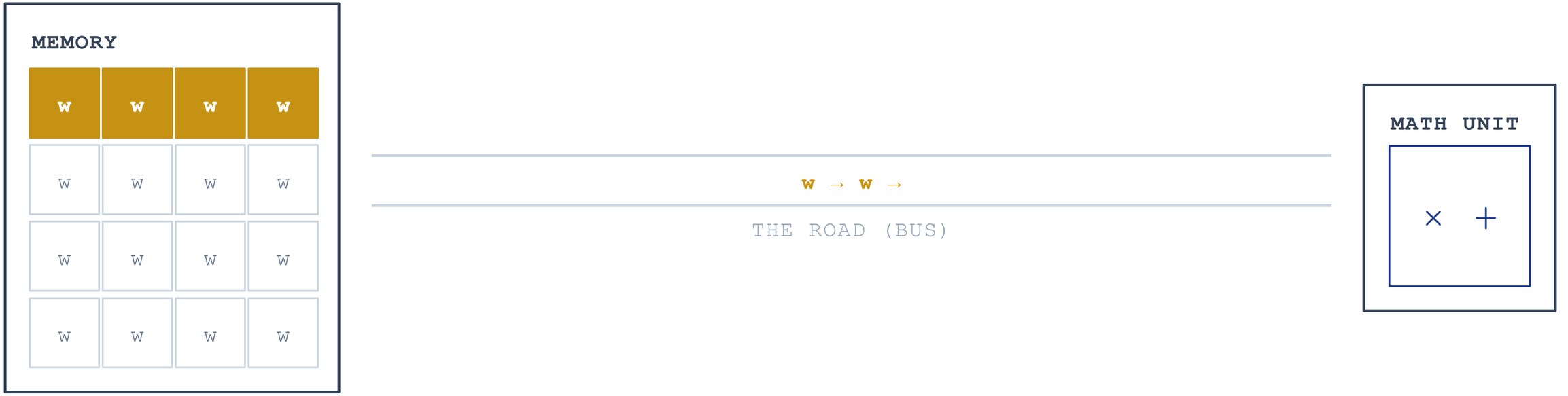
Fetch a weight. Carry it. Multiply. Send it back. That round trip already costs more than the multiply itself.

643.7 pJ

ONE FETCH + ONE MULTIPLY

Billions of trips make a traffic jam

STEP 3 OF 3 · THE TRAFFIC JAM



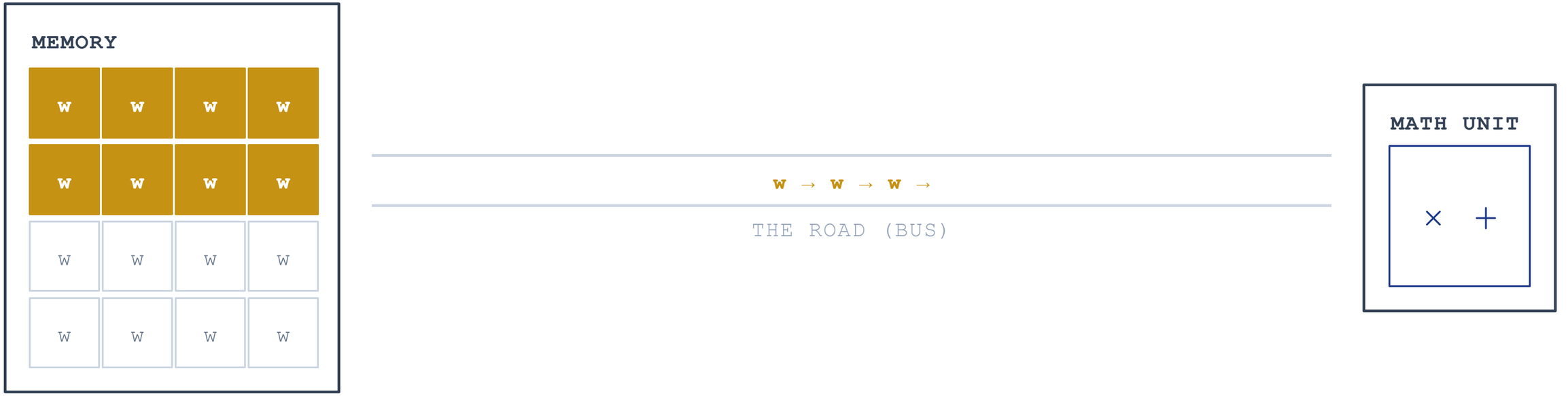
Every weight needs its own trip. The meter keeps running.

2,575 pJ

4 TRIPS · RUNNING ENERGY

Billions of trips make a traffic jam

STEP 3 OF 3 · THE TRAFFIC JAM



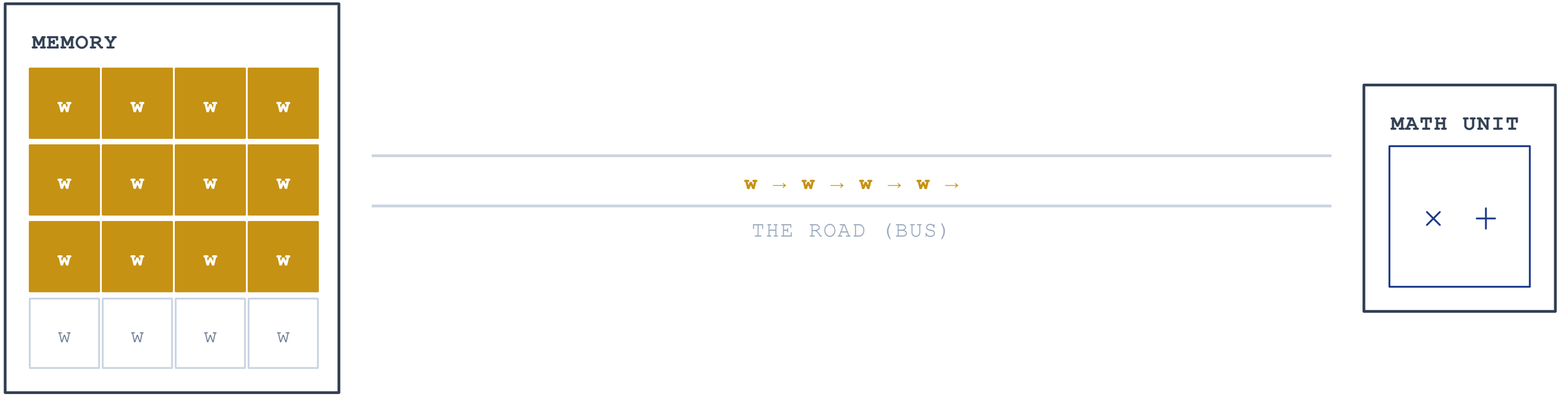
Every weight needs its own trip. The meter keeps running.

5,150 pJ

8 TRIPS · RUNNING ENERGY

Billions of trips make a traffic jam

STEP 3 OF 3 · THE TRAFFIC JAM



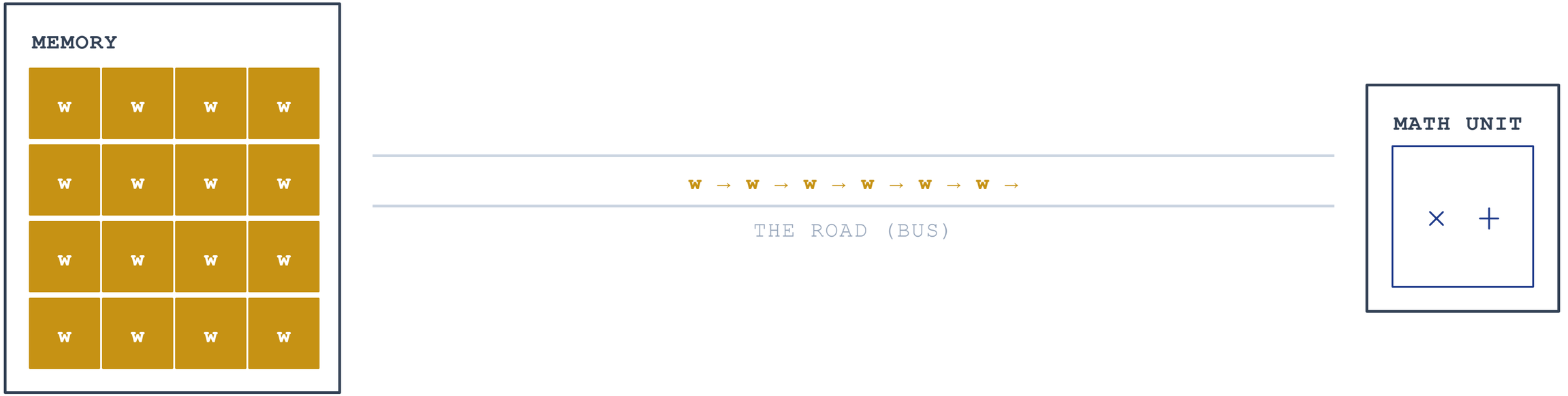
Every weight needs its own trip. The meter keeps running.

7,724 pJ

12 TRIPS · RUNNING ENERGY

Billions of trips make a traffic jam

STEP 3 OF 3 · THE TRAFFIC JAM



Hundreds of billions of weights per word, one road. Most of the time — and most of the energy — is spent in traffic.

10,299 pJ

16 TRIPS · RUNNING ENERGY

The trip costs 170 times more than the math

OPERATION	ENERGY
Fetch one weight from memory	640 pJ
Multiply-and-add that weight	3.7 pJ

Fetching costs ~173× more than computing. Almost all of the energy bill is the trip, not the math.

Scale it up: one chip

STEP 1 / 3

700 watts

ONE AI GPU, FLAT OUT

1 CHIP · ALWAYS ON

About half a microwave oven — except it never turns off.

Scale it up: one rack

STEP 2 / 3

10,000 watts

ONE SERVER RACK

8 CHIPS + EVERYTHING AROUND THEM

One rack draws what eight homes do. A datacenter holds thousands of racks.

Scale it up: one datacenter

STEP 3 / 3

100,000,000 watts

ONE AI DATACENTER ≈ 80,000 HOMES

A SMALL CITY, DRAINED INTO ONE BUILDING

A single AI datacenter draws the power of a small city — and hundreds more are being built right now.

AUDIENCE POLL

One ChatGPT answer uses how much more energy than a Google search?

A

About the same

B

~10×

C

~1,000×

Roughly ten times more

~3 Wh

ONE CHATGPT ANSWER

~0.3 Wh

ONE GOOGLE SEARCH

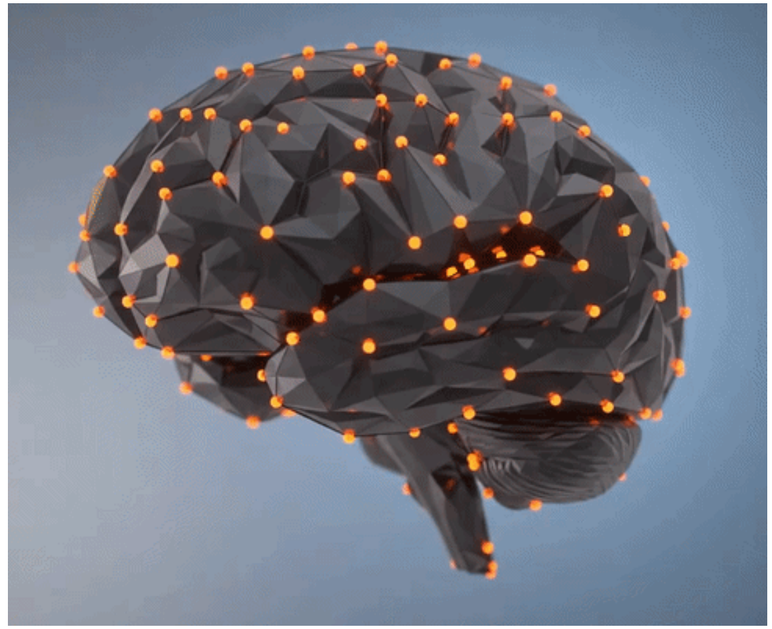
Datacenters already use ~1.5% of the world's electricity — on track to double by 2030.

IEA · 2024

SECTION 03

The 20-watt computer in your head

Nature already solved the energy problem.



Your brain runs on one dim lightbulb

20 watts

THE HUMAN BRAIN

700 watts

ONE AI CHIP

A single dim lightbulb runs the machine that learns language, recognizes every face you've met, and keeps your heart beating — and it still out-learns every datacenter on Earth.

Trick one: no road to travel

SYNAPSE

stores the weight AND multiplies with it — same place

A synapse never ships its weight anywhere. Memory and math are the same physical thing — no bus, no round trip.

Trick two: silence is free

NEURON

fires only when it has something to say

Most neurons are quiet most of the time. A chip that runs every circuit every clock cycle can't say that — the brain only spends energy on what matters, when it matters.

Trick three: slow, but unbelievably wide

86B
NEURONS

100T
SYNAPSES

Each neuron is slow — milliseconds, not nanoseconds. But 86 billion of them fire together, in parallel, all the time. Width beats speed.

Copying those tricks into silicon: neuromorphic engineering

01 Build memory and math into the same circuit — no road, no trip.

02 Only spend energy when there's a signal — event-driven, not clock-driven.

03 This field is called neuromorphic engineering — and it's what our lab does.

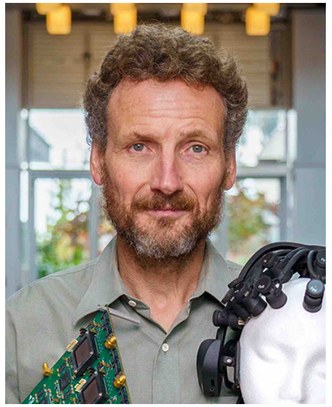
SECTION 04

What our lab builds

This is where I spend my days.

INTEGRATED SYSTEMS NEUROENGINEERING

Prof. Gert Cauwenberghs
UC San Diego



01

Micropower chips

Circuits inspired by brain architecture, built to run on almost nothing.

02

Neural interfaces

Hardware that talks directly to real neurons and neural systems.

03

My focus: compute-in-memory

Energy-efficient chip design that puts memory and math in the same place.

Sixteen weights, stored in a grid

STEP 1 OF 4 · THE SETUP



Just like before — except this memory has a hidden talent. It's called compute-in-memory.

Inputs flow in along the rows

STEP 2 OF 4 · INPUTS FLOW



Every cell in a row sees its input at the same moment. Nothing has been moved out of memory.

Inputs flow in along the rows

STEP 2 OF 4 · INPUTS FLOW



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Inputs flow in along the rows

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Inputs flow in along the rows

STEP 2 OF 4 · INPUTS FLOW



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Every cell multiplies at the same instant

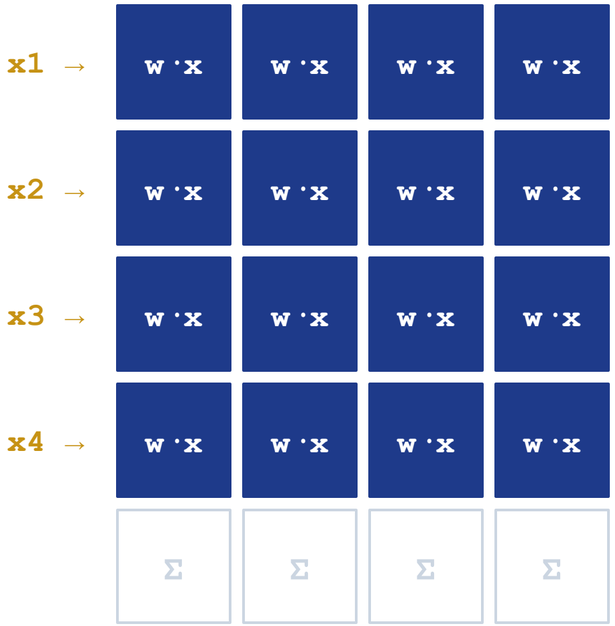
STEP 3 OF 4 · ALL MULTIPLY



Every row is live. Watch what happens the instant they all multiply.

Every cell multiplies at the same instant

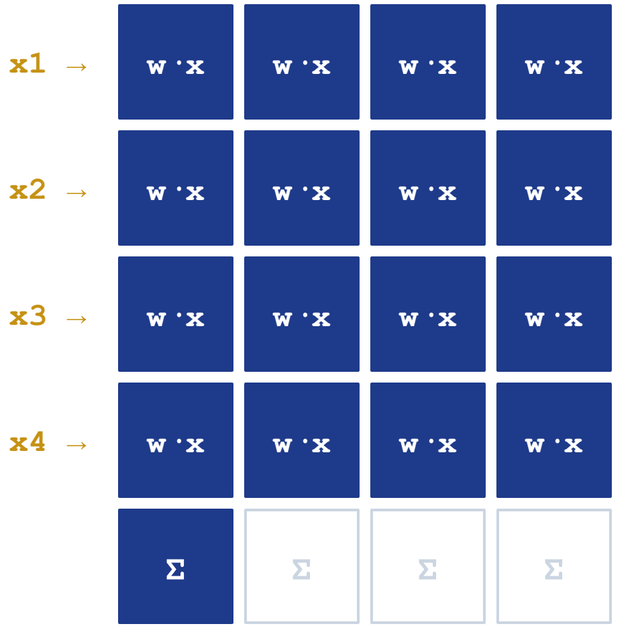
STEP 3 OF 4 · ALL MULTIPLY



Like a synapse. All sixteen multiplies happen at once, using the physics of the circuit itself.

Columns add it up — that's a matrix multiply

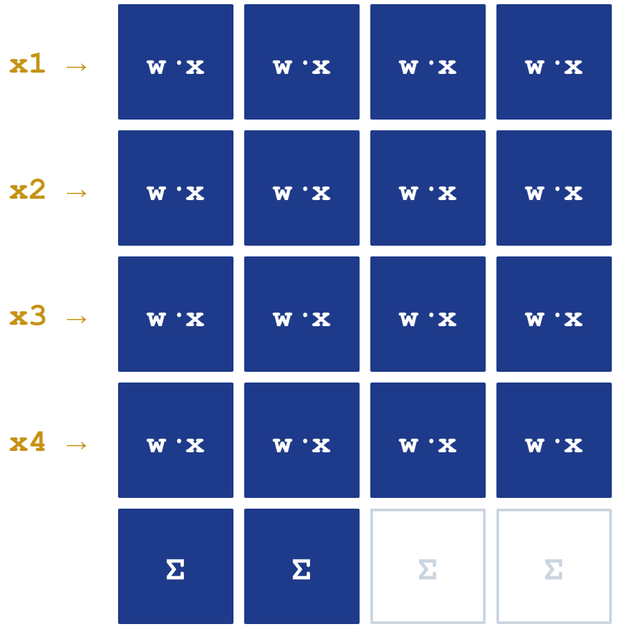
STEP 4 OF 4 · COLUMN SUM



Each column adds its products. That's a matrix multiply.

Columns add it up — that's a matrix multiply

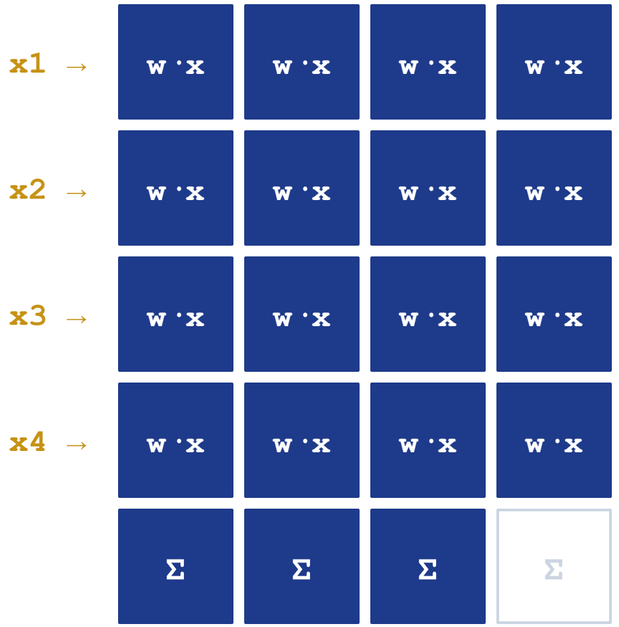
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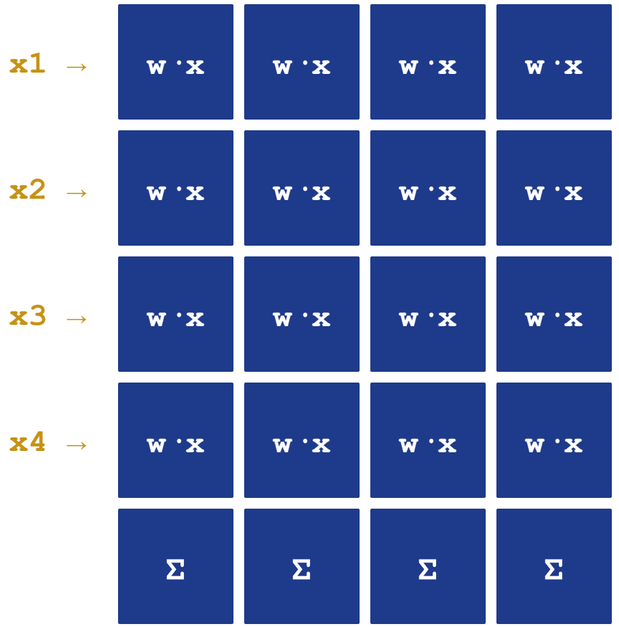
STEP 4 OF 4 · COLUMN SUM



Each column adds its products. That's a matrix multiply.

Columns add it up — that's a matrix multiply

STEP 4 OF 4 · COLUMN SUM



16 multiplies, 1 step, 0 round trips. The traffic jam never happens.

The race: shuttle vs. in-memory

THE SHUTTLE WAY

one weight fetched per trip

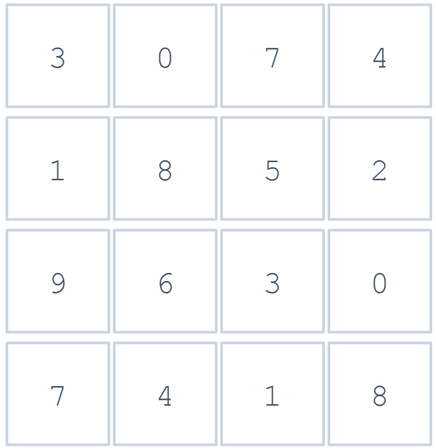


0
ENERGY · pJ

—
STATUS

THE IN-MEMORY WAY

every cell computes in place



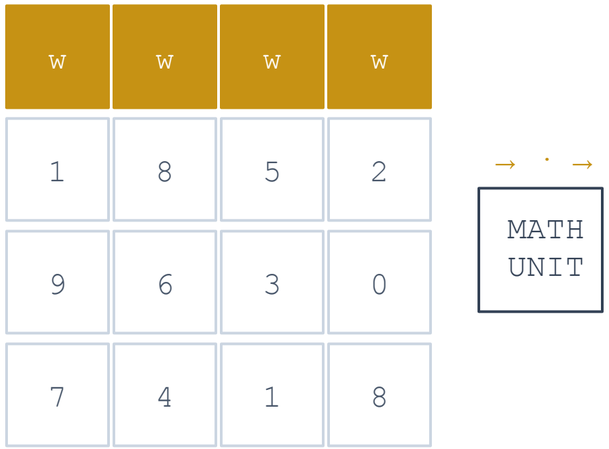
0
ENERGY · pJ

—
STATUS

The race: shuttle vs. in-memory

THE SHUTTLE WAY

one weight fetched per trip



2575
ENERGY · pJ

TRIP 4/16
STATUS

THE IN-MEMORY WAY

every cell computes in place



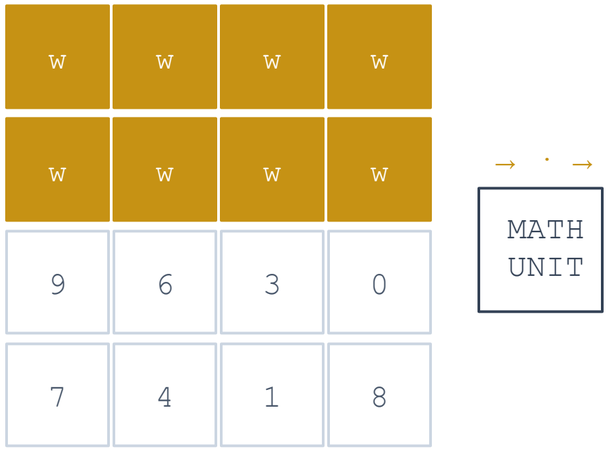
59
ENERGY · pJ

DONE
STATUS

The race: shuttle vs. in-memory

THE SHUTTLE WAY

one weight fetched per trip



5150
ENERGY · pJ

TRIP 8/16
STATUS

THE IN-MEMORY WAY

every cell computes in place



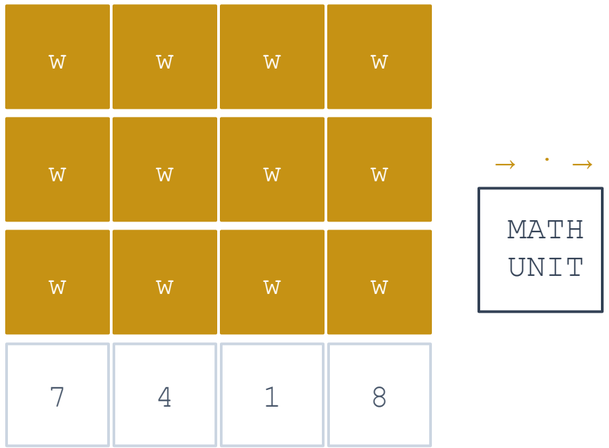
59
ENERGY · pJ

DONE
STATUS

The race: shuttle vs. in-memory

THE SHUTTLE WAY

one weight fetched per trip



7724
ENERGY · pJ

TRIP 12/16
STATUS

THE IN-MEMORY WAY

every cell computes in place



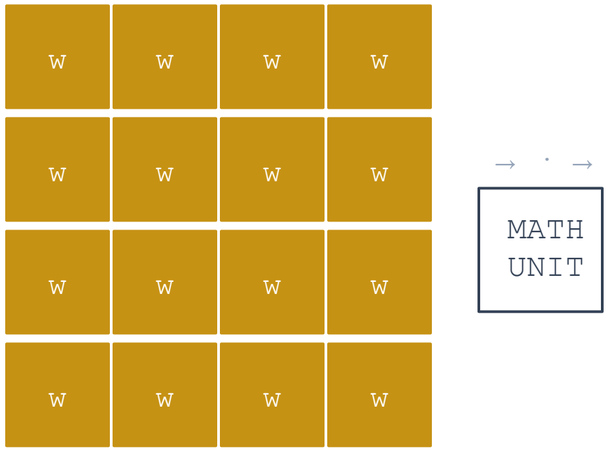
59
ENERGY · pJ

DONE
STATUS

The race: shuttle vs. in-memory

THE SHUTTLE WAY

one weight fetched per trip



10299
ENERGY · pJ

DONE
STATUS

THE IN-MEMORY WAY

every cell computes in place



59
ENERGY · pJ

DONE
STATUS

SECTION 05

Why this matters

What efficient hardware unlocks next.

AI that lives with you, not in a datacenter

01

Hearing aids that pick one voice out of a crowd

Neural networks running on a battery the size of a pea.

02

Wearables that catch problems before you feel them

Always-on health monitoring on-device, no need for large data-centers.

03

Implants that speak the brain's own language

Neural interfaces that must run for years on almost nothing — this is our lab's frontier.

Energy is the currency of next-generation AI

01 AI is just **simple** math at unbelievable scale — and the math is the cheap part.

02 The brain shows a better blueprint: compute where the memory lives, only when needed.

03 Whoever computes with the least energy will build the next generation of AI.

Questions

Ask about chips, brains, PhD life, or how to get started in hardware.

SHASHANK BANSAL · ISN.UCSD.EDU · UC SAN DIEGO